ITA 0502- COMPUTER VISION

LAB EXPERIMENT-15

15. Perform transformation using Direct Linear Transformation.

CODING:

```
import cv2
import numpy as np

# Read the image
img = cv2.imread("flower.png")

if img is None:
    print("Image not found!")
    exit()

rows, cols = img.shape[:2]

# Source points
src_points = np.float32([
    [50, 50],
    [300, 50],
    [50, 250],
    [300, 250]
])

# Destination points
dst_points = np.float32([
    [20, 80],
    [280, 30],
    [80, 280],
    [320, 250]
])
```

```
# DLT (Homography Estimation)
H, status = cv2.findHomography(src_points, dst_points, 0)

# Warp image
output = cv2.warpPerspective(img, H, (cols, rows))

# Display
cv2.imshow("Original Image", img)
cv2.imshow("DLT Transformation", output)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT: